Environmental Sound Classification Using Machine Learning

Machine Learning Internship — Technical Screening Task Report

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1. Objective

The objective of this project was to build a machine learning system that accepts an audio file and predicts its environmental sound category.

Four acoustically distinct classes were selected for the final model: dog bark, car horn, rain, and engine sound. These represent transient, harmonic, broadband, and low-frequency sound signatures respectively, giving a reasonable spread of acoustic characteristics for a proof-of-concept classifier.

2. Dataset Used

Dataset: ESC-50 (Environmental Sound Classification), a labeled collection of environmental audio recordings, with the four selected classes used from it.

- Classes used: dog bark, car horn, rain, engine sound (160 clips total, 40 per class).
- Train/test split: an 80% / 20% split was used (128 training clips, 32 testing clips). The original audio files were split before augmentation, to avoid data leakage between train and test.

3. Approach Followed

3.1 Feature Extraction

Thirteen Mel-Frequency Cepstral Coefficients (MFCCs) were extracted from each audio file using the Librosa library. MFCCs were selected because they provide a compact representation of the frequency characteristics of sound and are commonly used for audio-classification tasks. The MFCC values were averaged over time to create a fixed-size feature vector for every audio file.

3.2 Data Augmentation

The following augmentation techniques were applied only to the training data: Gaussian noise addition, time shifting using zero padding, pitch shifting, and time stretching. These methods increased the variety of the training samples and improved the model's ability to handle variations in audio.

3.3 Model Development

A Random Forest Classifier containing 200 decision trees was trained using the extracted MFCC features. Random Forest was selected because it works effectively with MFCC feature vectors, trains quickly on a normal laptop without requiring a GPU, provides feature-importance information, and is easy to explain during a technical demonstration.

4. Model Architecture

Component	Details
Feature extraction	13 MFCCs per clip (Librosa), mean-pooled across time frames
Feature vector size	13-dimensional
Classifier	Random Forest, 200 decision trees
Data augmentation	Gaussian noise, time shifting, pitch shifting, time stretching (training data only)
Train / test split	80% train (128 clips) / 20% test (32 clips), split before augmentation

5. Results Achieved

5.1 Overall Performance

Metric	Score
Accuracy	84.38%
Precision	87.78%
Recall	84.38%
F1-score	85.06%

5.2 Per-Class Performance

Class	F1-score
Car Horn	0.88
Dog Bark	0.93
Engine Sound	0.74
Rain	0.86

The confusion matrix showed that dog bark, car horn, and rain were classified effectively. Some samples were incorrectly classified as engine sound, which reduced the precision of the engine-sound class.

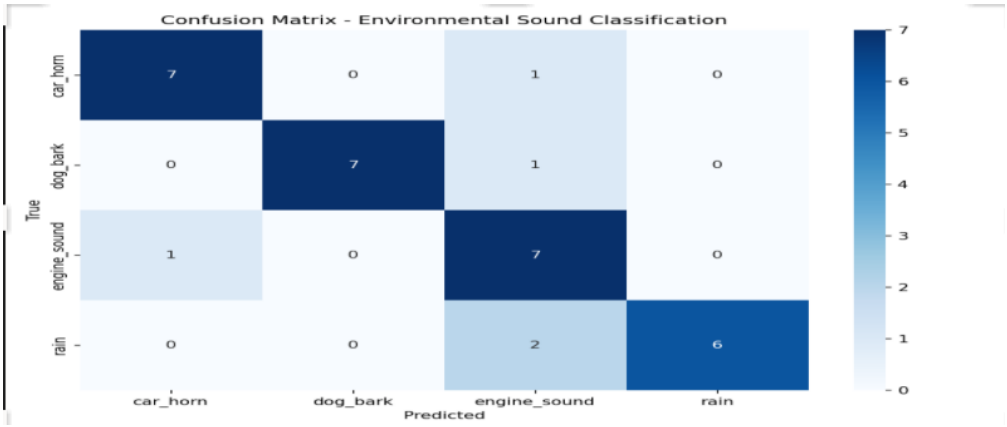


Figure 1 — Confusion matrix for the final four-class model.

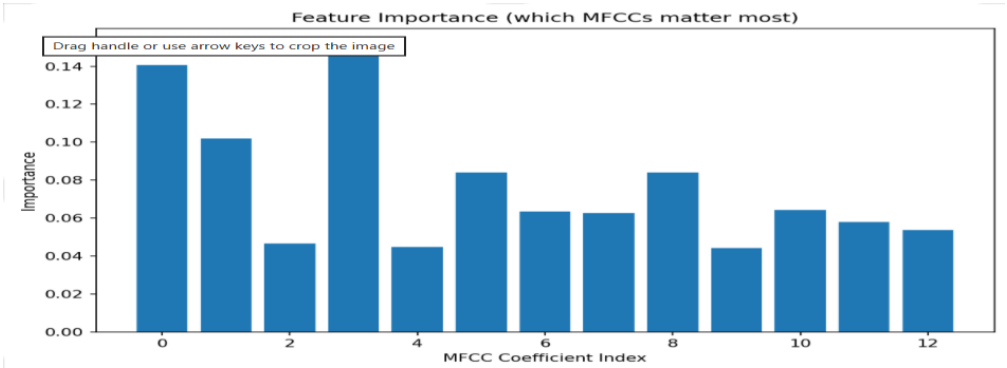


Figure 2 — MFCC feature importance from the Random Forest model.

6. Application and Testing

A Streamlit web application was developed to allow users to upload a WAV audio file. The application loads the trained model, extracts MFCC features, predicts the sound category, and displays confidence scores for all classes. The model was also tested using unseen audio files.

7. Challenges Faced

- Similar frequency patterns between some sound categories.
- Background noise in real-world recordings.
- Differences between ESC-50 samples and external recordings.
- Some dog-bark samples being confused with engine sounds.

8. Future Improvements

- Using a larger dataset.
- Training a CNN using Mel spectrogram images.
- Adding more sound categories.
- Improving noise robustness.
- Converting the model to TensorFlow Lite.
- Deploying the model on mobile or edge devices.

9. Conclusion

The project successfully developed an end-to-end environmental sound classification pipeline using MFCC features and a Random Forest model. The system includes dataset preparation, feature extraction, data augmentation, model training, evaluation, unseen-audio testing, and a Streamlit-based user interface.